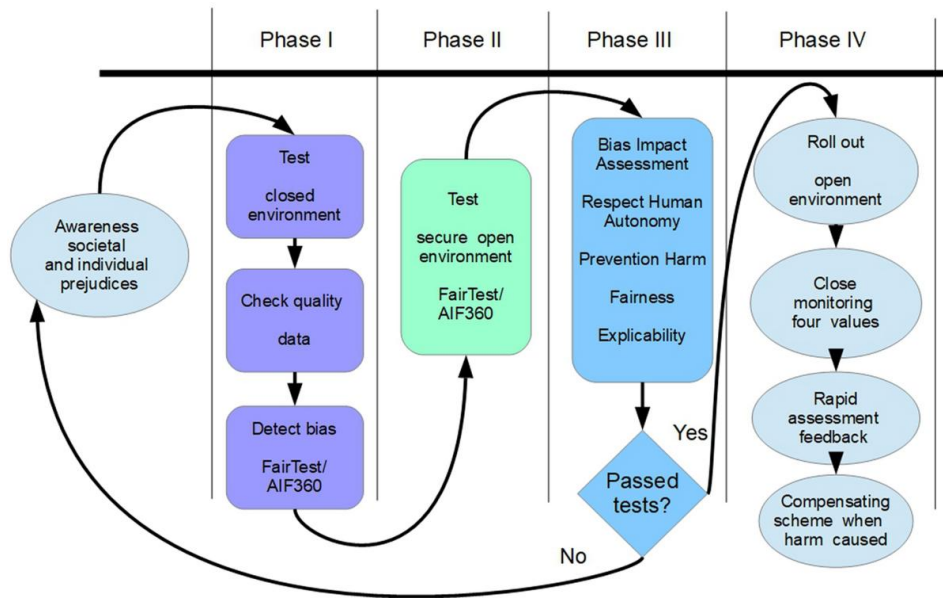
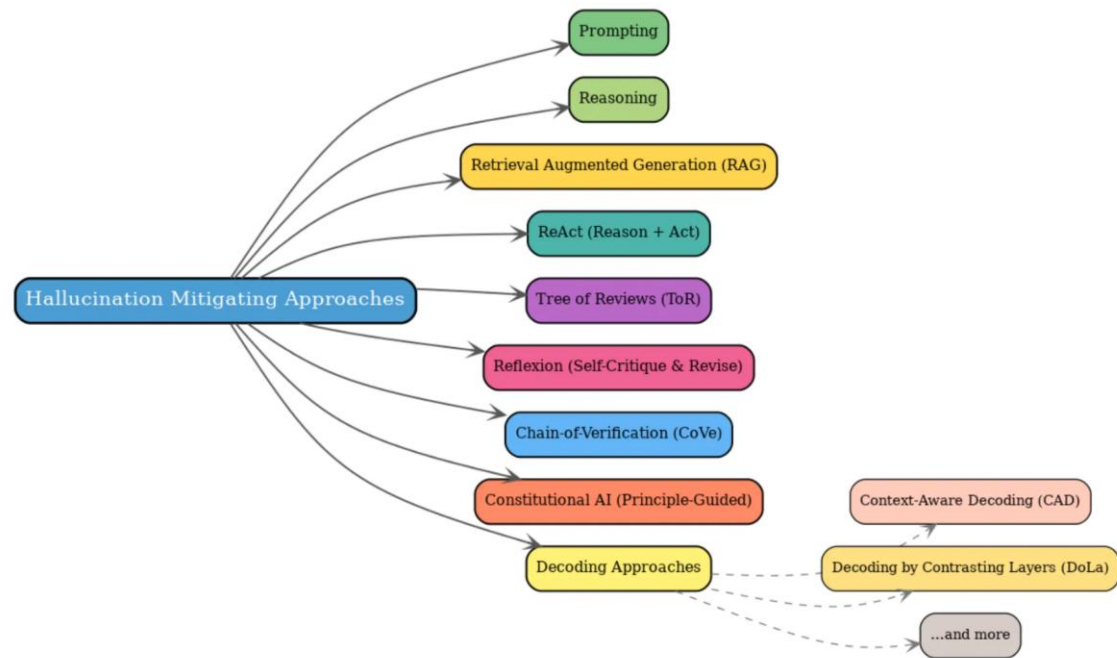

BIAS DETECTION: FIGHTING ALGORITHMIC UNFAIRNESS



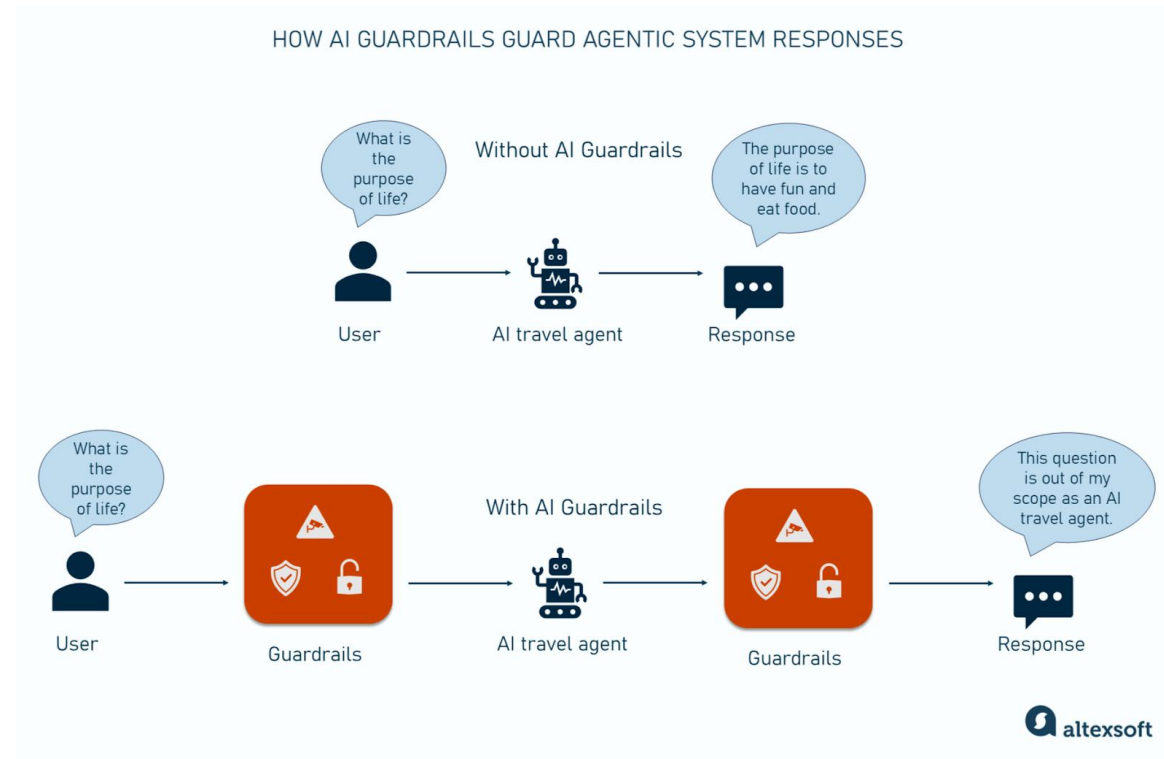
- **Algorithmic Bias:** Occurs when a model produces systematically prejudiced results (e.g., a medical AI that works better for one ethnic group than another).
 - **Data Reflection:** AI is a mirror; if the training data is biased, the model will be biased.
 - **Audit Strategies:** We must test models across different demographics (gender, age, location) to ensure "Equal Opportunity" and "Predictive Parity."
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HALLUCINATION MITIGATION: THE QUEST FOR TRUTH



- **Hallucination:** When an LLM confidently generates false information.
- **Root Cause:** LLMs are "stochastic parrots"—they predict the most likely word, not necessarily the most truthful one.
- **Mitigation Strategies:**
 - **RAG:** Grounding the model in external facts.
 - **Self-Consistency:** Asking the model to solve a problem multiple times and picking the consensus.
 - **Verification Loops:** Using a second "Judge" model to fact-check the first model's output.

DEPLOYMENT SAFETY: BUILDING THE GUARDRAILS



- **Harmful Content:** Preventing AI from generating instructions for illegal acts, hate speech, or self-harm.
- **Guardrails:** Software layers that sit between the user and the AI (e.g., Llama Guard or NeMo Guardrails).
- **Red Teaming:** The practice of intentionally trying to "break" the AI's safety rules to find vulnerabilities before the public does.